

PHYSICS PRACTICAL SHEETS

Date: 20/10/22

PN CAMPUS

Class: BSc 2nd year

Roll No.: 68

Shift: Day

Experiment No.: 9

Group: P2

Sub: Physics

Set: 1

Object of the Experiment (Block Letter):

OBJECT: TO PLOT A GRAPH BETWEEN CURRENT AND FREQUENCY IN A SERIES LCR CIRCUIT AND TO FIND THE RESONANT FREQUENCY.

APPARATUS REQUIRED:

(i) Function generator

(ii) LCR circuit device

(iii) Multimeter

(iv) Wires

THEORY:

The impedance of series LCR circuit is given by

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

It has minimum value when $\omega L = \frac{1}{\omega C}$. When this condition holds, the current in the circuit is maximum and this is called an acceptor circuit. The applied emf is in phase with the current. This condition is known as the resonant condition for the circuit and the frequency at resonance is called resonant frequency or natural frequency, which is equal to the frequency of emf applied. Therefore, if f_0 is the natural or resonant frequency then,

$$\omega L = \frac{1}{\omega C}$$

$$2\pi f_0 L = \frac{1}{2\pi f_0 C}$$

$$\text{or, } f_0 = \frac{1}{2\pi \sqrt{LC}}$$

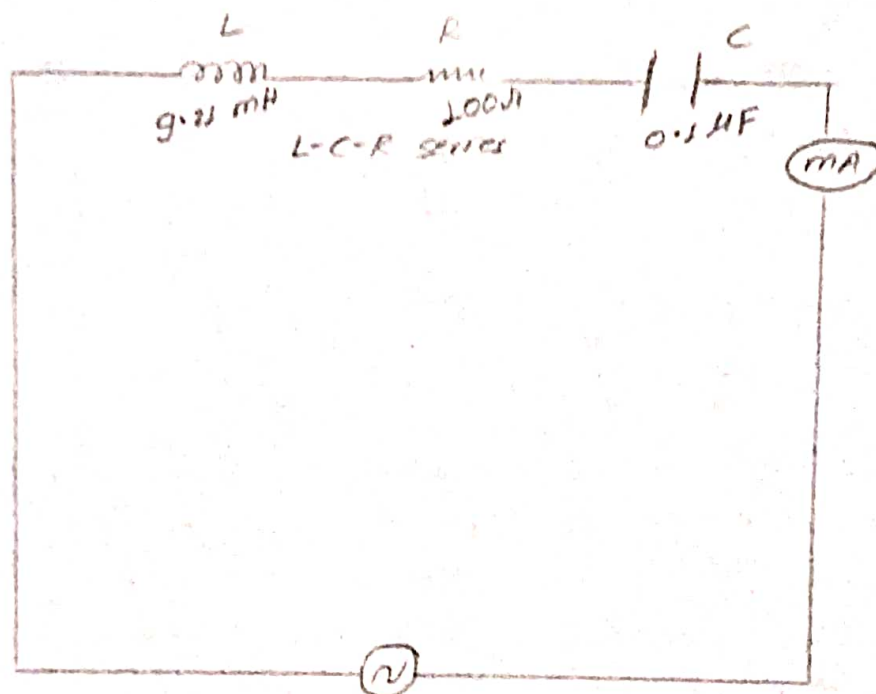


Fig: LCR circuit in series

At the frequency below and above the resonating frequency f_0 , the current in the circuit is less than the maximum value.

OBSERVATION:

10 divisions on the scale = 2 mA

1 division on the scale = $2/10$ mA

$\therefore$ L.C. of the device = 0.2 mA

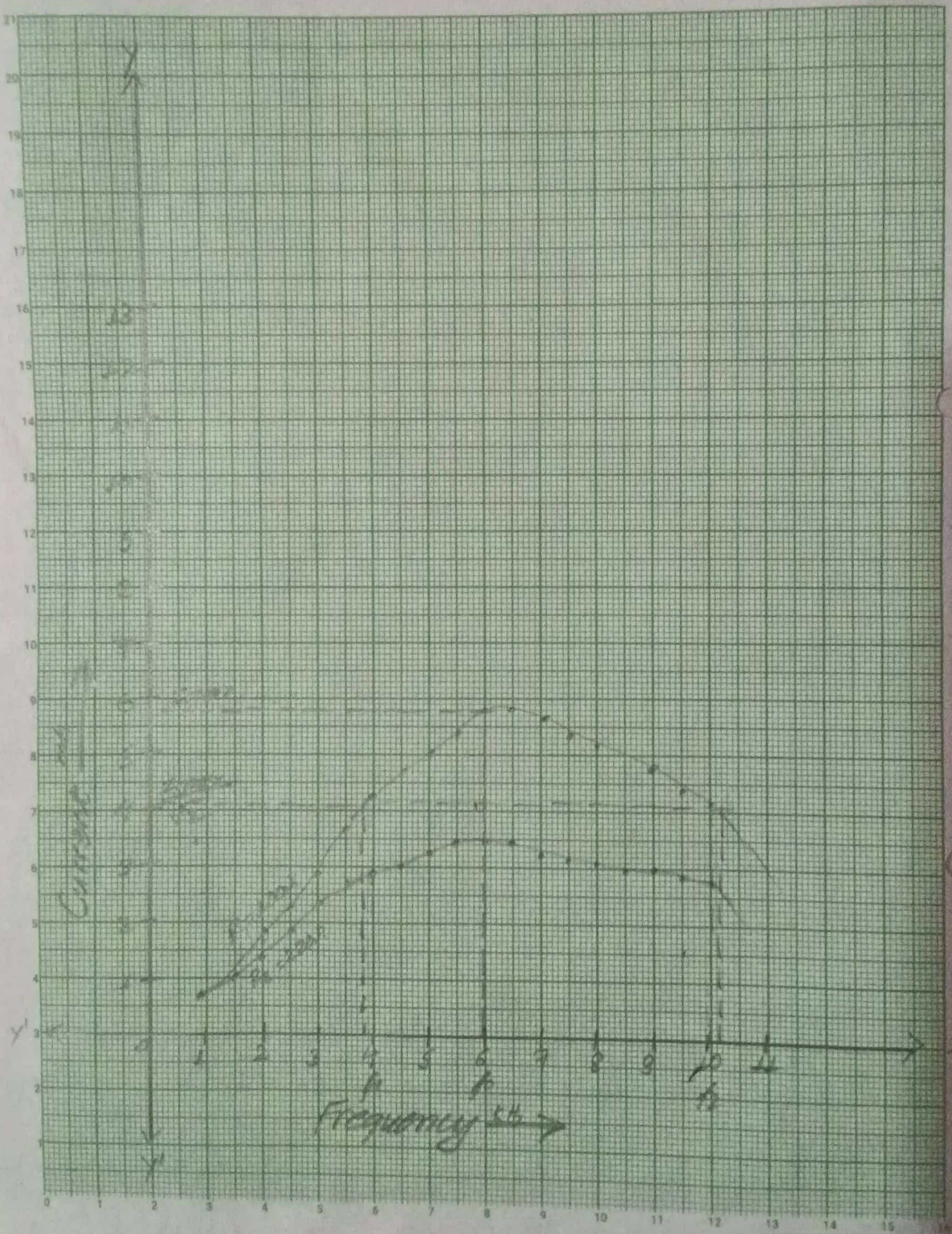
Inductance (L) = 9.21 mH = 9.21×10^{-3} H

Capacitance (C) = 0.1 μ F = 0.1×10^{-6} F

Resistance (R) = 200 Ω and 220 Ω

OBSERVATION TABLE:

S.N.	Frequency (KHz)	Current in the circuit in mA for	
		$R_1 = 100 \Omega$	$R_2 = 220 \Omega$
1.	1.0	0.6	0.6
2.	1.5	1	1
3.	2.0	1.8	1.4
4.	2.5	2.2	1.8
5.	3.0	2.8	2.4
6.	3.5	3.6	2.6
7.	4.0	4.2	2.8
8.	4.5	4.6	3
9.	5.0	5	3.2
10.	5.5	5.4	3.4
11.	6.0	5.8	3.4
12.	6.5	5.8	3.4
13.	7.0	5.6	3.2



14.	7.5	5.4	3.2
15.	8.0	5.2	3.1
16.	8.5	5	3
17.	9.0	4.8	3
18.	9.5	4.4	2.9
19.	10.0	4.2	2.8

CALCULATION:

$$\begin{aligned}
 \text{Resonance frequency, } f_0 &= \frac{1}{2\pi\sqrt{LC}} \\
 &= \frac{1}{2\pi\sqrt{9.22 \times 10^{-3} \times 0.1 \times 10^{-6}}} \\
 &= 5244.33 \text{ Hz} \\
 &= 5.244 \text{ kHz}
 \end{aligned}$$

From the graph the resonance frequency for $R=100\Omega$ is 5.8 kHz.

RESULT:

From the experiment, the resonance frequency of series LCR circuit was found to be 5.8 kHz.

ERROR ANALYSIS:

$$\begin{aligned}
 \text{percentage error} &= \left| \frac{\text{Standard value} - \text{Observed value}}{\text{Standard value}} \right| \times 100\% \\
 &= \left| \frac{5.244 - 5.8}{5.244} \right| \times 100\% \\
 &= 10\%
 \end{aligned}$$

CONCLUSION:

Hence, the resonance frequency of given series LCR circuit was found to be 5.8 kHz in the laboratory with 10 percent error.

PRECAUTIONS:

- Circuit should be well connected
- The value of inductance ' L ' and capacitance ' C ' should be properly selected.
- Current in the circuit should be noted carefully.
- The output voltage of oscillator should be kept constant.

Further,

$$\text{Quality factor (Q)} = 2\pi \left(\frac{f_0}{f_2 - f_1} \right) = 2\pi \left(\frac{5.8}{10.2 - 3.8} \right) \\ = 5.69$$

$$\text{Quality factor for inductance (Q)} = \frac{\omega L}{R_1} = \frac{2\pi f_0 L}{R_1} \\ = \frac{2\pi \times 5.8 \times 9.21 \times 10^{-3}}{100} \\ = 3.35 \times 10^{-3}$$

$$\text{Quality factor for capacitance (Q)} = \frac{X_C}{R_1} = \frac{1}{\omega C R_1} = \frac{1}{2\pi f_0 C R_1} \\ = \frac{1}{2\pi \times 5.8 \times 0.4 \times 10^{-6} \times 100} \\ = 2744.05$$

Ans,

$$\text{Band width} = (f_2 - f_1) \text{ Hz} \\ = (10.2 - 3.8) \text{ kHz} \\ = 6.4 \text{ kHz} \\ = 6400 \text{ Hz}$$

